

Nathaniel (Nate) Brewer

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Applied Software Engineering student (graduating May 2026) with hands-on experience in distributed systems, IoT/embedded hardware, full-stack web development, and AI-integrated applications. Published research co-author (IEEE Allot 2026, EIT 2026). Seeking enterprise software engineering roles in an Agile environment.

EDUCATION

B.S. Applied Software Engineering – Northern Kentucky University

Aug 2022 – May 2026

Expected Graduation: May 2026 | Dean's List: Fall 2024

Relevant Coursework

- Software Design – SOLID principles, APIEC patterns, OOP, GoF design patterns (Creational, Structural, Behavioral)
- Full-Stack Development – Node.js/MongoDB and PHP/MySQL web applications, RESTful API design, session and token-based authentication
- Cross-Platform Development – Flutter/Dart for Android, iOS, Web, Windows, macOS, and Linux from a single codebase
- Database Design – Relational schema design, SQL queries, NoSQL (MongoDB) document modeling

TECHNICAL SKILLS

Languages: Java, Python, C++, C, C#, Dart, Kotlin, JavaScript (Node.js), PHP

Frameworks & Runtimes: Flutter/Dart, Node.js, React (learning), Pygame

Web & Backend: Full-Stack, PHP/MySQL, Node.js/MongoDB, RESTful API design, JWT Authentication, Socket Programming

Databases: MySQL (relational schema, SQL), MongoDB (document modeling), JSON data interchange

Distributed Systems: TCP/IP socket programming, multi-threaded architectures, hierarchical edge computing, JSON packet protocols

Embedded & IoT: ESP32/Heltec LoRa32, Arduino IDE, LoRa/LoRaWAN, SD card logging, GPS/GNSS, CRC validation, packet fragmentation

AI / ML Tools: SepFormer (speech separation), Wav2Vec2 (audio embeddings), EnCodec (neural audio codec), Cohere API

AI Development Workflow: Spec-driven, human-in-the-loop development; structured prompt engineering; GitHub Copilot, Cursor, ChatGPT for design, planning, and implementation; GenAI as a technical research and learning accelerator

Build & DevOps: Maven, Git, GitHub (branching, merging, PR workflow, Actions CI/CD)

IDEs & Environments: VS Code, IntelliJ IDEA, Android Studio, Visual Studio Community, Arduino IDE

Operating Systems: Windows, Linux (Ubuntu/WSL), cross-platform scripting (Bash/PowerShell)

Support & IT: Help Desk, A/V Support, IT Assistance, Technical Documentation, Customer Service

EXPERIENCE

Undergraduate Research Assistant – Northern Kentucky University – School of Computing & Analytics May 2025 – Present

- Co-designing and implementing SpAudioX, a hierarchical three-layer edge computing framework for distributed spatial audio sensing; co-authored paper accepted to IEEE Allot 2026.
- Built a Java/Maven multi-threaded 3-tiered decentralized edge network (Coordinator → Server → Node) using TCP/IP sockets, JSON packet serialization (Gson), and Log4j2 structured logging.
- Integrated SepFormer speech separation and Wav2Vec2 embedding models into the edge server audio pipeline; implemented cross-stream consensus-based anchor identification and noise suppression.
- Deployed and evaluated the system on a multi-node virtual machine testbed (virtual LAN); voice-only SI-SDR improved ~4.2 dB and voice-with-music SI-SDR improved ~9.3 dB from Tier 1 to Tier 3.
- Developing a LoRa communication platform on ESP32/Heltec hardware (C++/Python); fragmented file transfer with CRC validation, ACK/timeout handling, SD card logging, and GPS/GNSS field diagnostics.
- Researching AI semantic voice compression via EnCodec neural codec achieving ~99.6% file size reduction (946 kB → 4 kB); paper accepted to EIT 2026.
- Collaborated with faculty advisor Dr. Rasib Khan on system design, architecture decisions, and testing strategy.

- Provided on-site A/V and IT support for 50+ university events including lectures, conferences, and academic ceremonies.
- Diagnosed and resolved hardware/software issues for faculty, staff, and students across university AV systems.
- Upgraded and maintained technical documentation processes, improving knowledge transfer and reducing repeat support tickets.

- Served in various roles across 6 years, including just under a year as Team Manager leading a team of 8 multi-shift employees.
- Managed inventory tracking and ordering across 500+ products, maintaining stock accuracy and minimizing shrink.
- Developed team communication and shift-handoff protocols, improving day-to-day operational efficiency.

RESEARCH & PUBLICATIONS

SpAudioX: A Hierarchical Edge Framework for Spatial Audio Sensing and Monitoring Applications

Nathaniel Brewer & Rasib Khan | Northern Kentucky University | IEEE AllIoT 2026

- Proposed a three-layer edge computing architecture (edge nodes → edge servers → edge coordinator) for distributed audio sensing without reliance on centralized cloud infrastructure.
- Designed spatially aware multi-source audio fusion using anchor-based stream alignment, source frequency scoring, and AI-driven signal enhancement at the edge server layer.
- Implemented fully functional Java prototype deployed across multiple VMs on a virtual LAN; evaluated using SI-SDR and Mel cosine similarity metrics across three experimental tiers.
- Targeted applications: disaster response, remote environmental monitoring, dynamic security environments.

Semantic Voice Compression for LoRa Mesh Networks in Infrastructure-Less Environments

Nathaniel Brewer et al. | EIT 2026

- Demonstrated ~99.6% audio compression using the EnCodec neural audio codec (946 kB → 4 kB; 7,391 packets reduced to 32), enabling voice transmission within LoRa's 11–242 byte/packet bandwidth constraint.
- Proposed evaluation of a generative neural vocoder for on-device speech reconstruction across rural, urban, and mountainous environments; metrics: latency, packet loss, semantic similarity, MOS, and energy efficiency.

PROJECTS

AI Audio Enhancement Research | Python, Flutter/Dart, C++, CMake | github.com/brewern5/AI-Audio-Enhancement-Research

- Foundational research project that seeded the LoRa semantic compression work — investigating whether AI can enhance lossy audio to perceptual lossless quality natively on streaming-capable devices, and whether this approach could work around Bluetooth's hard exclusion of lossless audio.
- Built a mock audio streaming API using a Python backend (backend/) with AI model integration (ai-models/) to process and serve enhanced audio, paired with a Flutter/Dart frontend (frontend/) for cross-platform display and playback.
- Structured as a full research repository: experiments/, data/, scripts/, docs/, and links/ directories supporting reproducible investigation and iterative model evaluation.
- Spans Dart, Python, C++, and CMake — reflecting the cross-layer nature of the problem from ML inference to platform-native audio handling.

DecentralizedEdgeNetwork | Java, Maven, TCP/IP, JSON, Log4j2 | github.com/brewern5/DecentralizedEdgeNetwork

- Built a Maven-managed 3-tiered decentralized edge network (Coordinator / Server / Node) as independent Java 17 processes communicating over TCP/IP sockets with JSON packet serialization (Gson) and Log4j2 structured logging.
- Performed a significant refactor that reduced the codebase to approximately one-third of its original size while maintaining all existing functionality and simultaneously adding new capabilities — a deliberate exercise in cohesion, clarity, and eliminating redundancy.

- Refactoring goal was architectural: introduced well-defined extension points throughout so future features (authentication, encryption, load balancing, edge intelligence) can be integrated without restructuring existing components — applying Open/Closed and Dependency Inversion principles at the system level.
- Designed a JSON packet protocol supporting 10+ message types (INIT, AUTH, MESSAGE, HEARTBEAT, STATUS, ACK, DISCONNECT, etc.) with a shared core/ layer enabling consistent packet handling across all tiers.
- Multi-threaded connection handling with keep-alive timers, cluster ID assignment, and full connection lifecycle management; hierarchical package layout (coordinator.*, server.*, node.*, core.*).

LoRa Research Platform | C++, Python, ESP32, Arduino IDE | github.com/brewern5/LoRa-Research

- Designed and implemented point-to-point LoRa communication on ESP32/Heltec LoRa-32 hardware with custom packet structures, CRC validation, and true half-duplex ACK/timeout handling.
- Built a fragmented file transfer protocol supporting audio and log file transmission under LoRa's low-bandwidth constraint, including packet sequencing and reassembly.
- Added SD card logging, on-device OLED status display, and GPS/GNSS data capture for real-world range and mobility field testing.
- Evolving toward a 3+ node mesh network (mesh/ module) to measure per-hop latency, spread factor effects, and packet delivery reliability; integrating EnCodec AI compression for semantic voice transmission.

SpAudioX Audio Processing Pipeline | Java, Python, SepFormer, Wav2Vec2, FFMPEG

- Integrated SepFormer (SpeechBrain WHAMR 16 kHz) to decompose mixed audio inputs into two candidate source signals per stream.
- Used Wav2Vec2 embeddings for cross-stream similarity scoring; implemented consensus-based anchor identification selecting the most persistent source across all mixtures.
- Generated spatially simulated audio mixtures via FFMPEG with 3 gain configurations (75/25, 50/50, 25/75) across 3 experimental tiers; all evaluation metrics stored in JSON for reproducibility.

Netflix – Cross-Platform Streaming App | Flutter, Dart, Node.js, GitHub Actions | github.com/Cross-platform-dev-456/netflix

- Team project: built a Netflix-inspired application in Flutter/Dart targeting Android, iOS, Web, Windows, macOS, and Linux from a single codebase.
- Implemented Home screen (genre rows, hero banner), Search screen (grid display), and Details screen (episode list, season selector); integrated a Node.js backend server for content delivery.
- Set up GitHub Actions CI/CD pipeline and maintained integration tests, team coding standards, and PR-based code review workflow.

Craigslist-Style Marketplace CMS | Node.js, MongoDB, Cohere API, JWT | github.com/EmmanNHop/ASE-220-MidtermProject

- Built a full-stack classified-listing web application using Node.js and MongoDB with a document-modeled schema for listings and users.
- Implemented JWT-based user authentication and authorization for protected routes.
- Integrated Cohere's generative AI API with a custom prompt to automatically generate relevant tags from user post content at submission — a production-style GenAI feature integration.

Music Review & Sharing Platform | PHP, MySQL, CSS | github.com/brewern5/ASE230-FinalProject

- Built a music review and discussion platform where users can post song reviews and other users can comment; includes admin and standard user roles.
- Designed relational MySQL schema with entities for users, posts, and comments; implemented full sign-up/sign-in/sign-out authentication flow in PHP.
- Structured with MVC-style separation: entity/ classes, db.php, dbfunctions.php, and individual page controllers (post.php, details.php, index.php).

ADP – Accelerated Debt Payment App | Flutter, Dart | github.com/brewern5/ADP---Accelerated-Debt-Payment

- Cross-platform Flutter/Dart app implementing the debt snowball method to generate optimized payoff plans from user-entered loan data (principal, interest rate, payment, term).

- Features dynamic amortization schedule generation, push notifications for payment reminders, and a payment confirmation flow to keep schedules current.
- Handles both fixed and variable interest rates; calculates optimal payoff order to minimize total interest paid.

Tetris – Software Design Principles Project | Python, Pygame | github.com/DannyPoff31/ASE420-Tetris

- Team software design exercise using Tetris as the implementation vehicle — the primary goal was deliberate application of SOLID principles and OOP design patterns, with the game itself secondary to the architectural exercise.
- Planned architecture upfront as a team with extensibility as a first-class concern: applied Open/Closed principle so new piece types and rules could be added without modifying existing logic; enforced Single Responsibility across game loop, board state, piece behavior, and rendering components.
- Discussed and documented design tradeoffs during planning; decisions were made with future maintainability in mind, mirroring an enterprise feature design and review process.
- Managed collaborative development via Git branching and PR-based code review; configured cross-platform virtual environment scripts for Windows (PowerShell) and Linux/WSL (Bash).

TuxGuitar – Open Source Contribution | Java | github.com/brewern5/tuxguitar-CopyToEndFeature

- Contributed a “Copy to End” feature to TuxGuitar, an open-source Java multitrack tablature editor with 3,800+ commit history.
- Navigated a large existing Java/SWT codebase to identify the relevant copy modal logic and extend it with new UI functionality; submitted via pull request against the upstream repository.

AI DEVELOPMENT APPROACH & TOOLING

Consistent practitioner of spec-driven, human-in-the-loop AI-assisted development:

- Define and document a feature/requirements list before any implementation begins — scoping what needs to be built, edge cases, and acceptance criteria.
- Review and iteratively refine requirements through discussion (with teammates or solo deliberation), ensuring clarity and completeness before involving AI tools.
- Construct a structured implementation plan specifically designed for the AI model: ordered, unambiguous steps that can be executed reliably by tools like GitHub Copilot or Cursor.
- Guide the model step-by-step through the plan, reviewing each output, catching misalignments, and correcting course — treating AI as a collaborative executor rather than an autonomous agent.
- Apply the same structured approach to technical learning: use GenAI tools as interactive research assistants to explore unfamiliar libraries, architectures, and concepts — forming precise questions, evaluating responses critically, and building mental models before writing code.
- Result: faster feature delivery with human-level review quality maintained throughout; consistent, well-planned codebases rather than ad-hoc AI-generated output.

ADDITIONAL INFORMATION

- Dean’s List, Northern Kentucky University – Fall 2024
- Research published at IEEE AllIoT 2026 (SpAudioX) and EIT 2026 (Semantic Voice Compression for LoRa).
- Agile collaboration experience: Git-based code review, branching strategies, and multi-person sprint planning on academic team projects.
- References and full GitHub project links available upon request.